Design and Development of a Real-Time WDM Optical Network Monitoring and Performance Analysis System

Configuration Reference: All Parameters, Options and Allowed Values

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Subject area: Electronics and Communication Engineering (Fibre Optical Communication)

About this document

The earlier summary listed only the default value of each setting. This reference lists every setting of the simulator together with all the values it can take: the full list of choices for drop-down and switch controls, and the minimum, maximum and step for numeric controls. The defaults are repeated in the last column.

How to read the tables:

- Parameter: the name shown in the dashboard. The configuration class that stores it (FiberConfig, WDMConfig, TxRxConfig, NoiseConfig, AmpConfig, DispConfig, TrafficConfig, FaultConfig) is given in the section title.
- All possible values: for a choice list, every option; for a number, the range allowed in the dashboard written as minimum to maximum with the step size. Values outside the range are not accepted by the dashboard. If a setting is changed in a saved JSON file instead, the simulator checks it and may show an error or a warning.
- Default: the value when the dashboard starts or after pressing Reset to Defaults. Some defaults change automatically when a related choice changes, for example the core size changes with the fibre material, structure and profile. This is stated in the table.
- Units: dB = decibel, dBm = decibel relative to 1 mW, km = kilometre, nm = nanometre, GHz = gigahertz, Gb/s = gigabit per second, ps = picosecond, ms = millisecond, us = microsecond.

All values describe a simplified simulation. Numbers marked as assumed in the model are representative engineering values and are not measurements of real hardware.

1. Fibre configuration (FiberConfig)

Table 1. Fibre parameters

Parameter	What it controls	All possible values (dashboard limits)	Default
Material	Type of fibre. Sets the attenuation curve, the dispersion model, the usable wavelength window and the default geometry.	Silica (glass); POF (plastic optical fibre, PMMA); PCS (plastic-clad silica).	Silica
Structure (mode)	Single-mode or multimode propagation. Changes the default core size and decides which dispersion terms are used.	Single-mode; Multimode.	Single-mode

Parameter	What it controls	All possible values (dashboard limits)	Default
Refractive-index profile	Step-index or graded-index core. A graded-index core greatly reduces modal dispersion in multimode fibre.	Step-index; Graded-index.	Step-index
Fibre length (km)	Link length. The slider and the numeric box always show the same value.	Slider: 1 to 100 km, step 1. Numeric box: 0.1 to 400 km, step 1.	20 km
Core diameter (um)	Diameter of the light-guiding core. Used for the numerical aperture check, the V-number and the mode count.	1 to 2000 um, step 0.5.	8.2 um for silica single-mode. Changes with material, structure and profile (see Table 2).
Cladding diameter (um)	Outer diameter of the cladding. It must be larger than the core.	2 to 3000 um, step 1. A value not larger than the core gives an error.	125 um for silica single-mode. Changes with the fibre type.
Core refractive index	Index of the core. Must be larger than the cladding index for the light to be guided.	1.2000 to 2.0000, step 0.0001.	1.4682 for silica single-mode. Changes with the fibre type.
Cladding refractive index	Index of the cladding.	1.2000 to 2.0000, step 0.0001. A value equal to or above the core index gives an error.	1.4633 for silica single-mode. Changes with the fibre type.
Numerical aperture (calculated)	NA = square root of (n_core squared minus n_clad squared). It is calculated, not typed in.	Result of the two indices. A value above 0.6 gives a warning.	About 0.12 for the default silica single-mode fibre.
V-number and mode count (calculated)	V = 2 x pi x core radius x NA / wavelength. A step-index fibre is single-mode when V is 2.405 or less.	Result of the geometry and the wavelength. The label (single-mode or multimode) is checked against V and a warning is shown if they disagree.	V is about 2.0 at 1550 nm for the default fibre.

Parameter	What it controls	All possible values (dashboard limits)	Default
Operating wavelength follows the WDM channels	When on, the operating wavelength is the average of the channel wavelengths from the WDM tab.	On; Off.	On
Operating wavelength (nm)	Wavelength used for the V-number, the cut-off check and as the reference point of the manual attenuation and dispersion values.	400 to 1700 nm, step 10. Editable only when the follow switch is off.	About 1550.12 nm (average of the default channels).
Manual attenuation	Optional replacement of the model attenuation. The whole attenuation curve is scaled so that it equals the typed value at the operating wavelength.	Switch: Off; On. When on: 0.01 to 50000 dB/km, step 0.01.	Off (model value, about 0.19 dB/km for silica single-mode at 1550 nm)
Manual chromatic dispersion	Optional replacement of the model dispersion D at the operating wavelength. The whole curve is shifted.	Switch: Off; On. When on: -1000 to 1000 ps/(nm km), step 0.5.	Off (model value, about 17.3 ps/(nm km) for silica single-mode at 1550 nm)
PMD coefficient (ps per square root km)	Polarisation-mode dispersion. Used only when the fibre is physically single-mode.	0 to 2, step 0.01.	0.1 for silica single-mode; 0 for the multimode presets.
Modal bandwidth (MHz km)	Effective modal bandwidth of a graded-index multimode fibre. A value of 0 means an ideal parabolic profile is assumed (optimistic).	0 to 20000 MHz km, step 50.	0 for single-mode and step-index fibres; 500 silica graded-index; 150 POF graded-index; 100 PCS graded-index.
Connectors (count)	Number of connectors in the link. Each adds the connector loss.	0 to 20, step 1.	2
Loss per connector (dB)	Insertion loss of one connector (assumed value).	0 to 3 dB, step 0.1.	0.5 dB
Splice loss (dB)	Loss of one splice (assumed value).	0 to 1 dB, step 0.01.	0.05 dB
Splice spacing (km)	Distance between splices. Number of splices = whole number of (length / spacing). Not shown in the dashboard; set it in a JSON configuration.	Any positive number of km. A value of 0 or below means no splices.	2 km

Parameter	What it controls	All possible values (dashboard limits)	Default
Extra margin (dB)	Additional fixed loss added to the link budget.	0 to 10 dB, step 0.5.	0 dB

1.1 All twelve fibre combinations

Material (3) x structure (2) x profile (2) gives twelve combinations. Choosing any of them fills in the defaults below. The V-number and the attenuation are evaluated at the default operating wavelength of that material (650 nm for POF, 850 nm for PCS and for multimode silica, 1550 nm for single-mode silica). If the V-number does not match the label, the label is flagged with a warning.

Table 2. Defaults for every fibre combination

Material	Structure	Profile	Core / cladding (um)	n core / n clad	NA	V	Behaves as	Default wavelength (nm)	Attenuation (dB/km) and D (ps/(nm km))	Status
Silica	Single-mode	Step-index	8.2 / 125	1.4682 / 1.4633	0.12	1.99	single-mode	1550	0.19 and 17.3	Valid
Silica	Single-mode	Graded-index	8.2 / 125	1.4682 / 1.4633	0.12	1.99	single-mode	1550	0.19 and 17.3	Valid, warning shown
Silica	Multimode	Step-index	50 / 125	1.48 / 1.4636	0.22	40.6	multimode	850	2.53 and -97.6	Valid
Silica	Multimode	Graded-index	50 / 125	1.48 / 1.4665	0.199	36.86	multimode	850	2.53 and -97.6	Valid
POF	Single-mode	Step-index	6 / 125	1.492 / 1.4898	0.081	2.35	single-mode	650	150 and -300	Unusual combination, warning shown
POF	Single-mode	Graded-index	6 / 125	1.492 / 1.4898	0.081	2.35	single-mode	650	150 and -300	Unusual combination, warning shown
POF	Multimode	Step-index	980 / 1000	1.492 / 1.402	0.51	2417.31	multimode	650	150 and -300	Valid
POF	Multimode	Graded-index	500 / 750	1.492 / 1.478	0.204	492.78	multimode	650	150 and -300	Valid

Material	Structure	Profile	Core / cladding (um)	n core / n clad	NA	V	Behaves as	Default wavelength (nm)	Attenuation (dB/km) and D (ps/(nm km))	Status
PCS	Single-mode	Step-index	8.2 / 125	1.4682 / 1.4633	0.12	3.63	multimode	850	5.42 and -91.8	Unusual combination, warning shown
PCS	Single-mode	Graded-index	8.2 / 125	1.4682 / 1.4633	0.12	3.63	multimode	850	5.42 and -91.8	Unusual combination, warning shown
PCS	Multimode	Step-index	200 / 230	1.457 / 1.4105	0.365	269.92	multimode	850	5.42 and -90.7	Valid
PCS	Multimode	Graded-index	200 / 230	1.457 / 1.4105	0.365	269.92	multimode	850	5.42 and -90.7	Valid, warning shown

Notes on the unusual combinations

- Silica, single-mode, graded-index: graded-index profiles are not used in ordinary single-mode fibre. The V-number and dispersion are calculated as for a step-index core.
- POF, single-mode (step or graded): single-mode plastic fibre is not a commercial product. The values are illustrative only.
- PCS, single-mode (step or graded): plastic-clad silica has a large core. At 850 nm the default 8.2 um core gives V above 2.405, so the fibre is physically multimode and modal dispersion is applied, together with the warning.
- PCS, multimode, graded-index: not a standard product. The modal bandwidth of 100 MHz km is an assumed value.
- Any combination can be modified afterwards by editing the core, cladding and index values. The dashboard then re-checks V, the mode label and the index order.

1.2 Attenuation and dispersion reference values of the model

Table 3. Model attenuation and chromatic dispersion at standard wavelengths

Fibre (material, structure, profile)	Wavelength (nm)	Attenuation (dB/km)	Chromatic dispersion D (ps/(nm km))
Silica single-mode step-index	850	1.48	-91.8
Silica single-mode step-index	1310	0.30	-0.3
Silica single-mode step-index	1383	0.37 (water absorption peak)	6.0
Silica single-mode step-index	1550	0.19	17.3
Silica single-mode step-index	1625	0.23	21.4
Silica multimode graded-index	850	2.53	-97.6
Silica multimode graded-index	1300	0.50	-2.9
PCS multimode step-index	850	5.42	-90.7
PCS multimode step-index	1300	3.85	-0.9
POF multimode step-index	520	100	-300 (constant, assumed)
POF multimode step-index	570	90	-300 (constant, assumed)
POF multimode step-index	650	150	-300 (constant, assumed)
POF multimode step-index	780	about 2080	-300 (constant, assumed)

Usable wavelength windows of the models: silica 600 to 1700 nm, PCS 600 to 1650 nm, POF 450 to 800 nm. Channels outside the window produce a warning. For POF above 850 nm the attenuation is set to 20000 dB/km (the fibre is effectively opaque).

2. WDM configuration (WDMConfig)

Table 4. WDM parameters

Parameter	What it controls	All possible values (dashboard limits)	Default
Scheme	Type of wavelength grid.	DWDM (dense, frequency grid); CWDM (coarse, 20 nm grid).	DWDM
Number of channels	How many wavelengths are multiplexed.	1 to 16 (slider, step 1). The CWDM standard allows 18, the dashboard limits it to 16.	8
Central wavelength (nm), DWDM	Centre of the DWDM grid. Channels are placed symmetrically around it on a fixed frequency spacing.	1200 to 1700 nm, step 0.1. Greyed out while custom wavelengths are on.	1550.12 nm (193.4 THz)
Frequency spacing (GHz), DWDM	Spacing between neighbouring channels in frequency.	12.5; 25; 50; 100; 200 GHz (the ITU-T G.694.1 values). At 1550 nm, 100 GHz is about 0.8 nm.	100 GHz
Start wavelength (nm), CWDM	First (shortest) CWDM channel.	1260 to 1625 nm, step 1. The ITU grid is 1271 + 20 k nm. Other values give an information message.	1471 nm
Spacing (nm), CWDM	Spacing between neighbouring CWDM channels.	2 to 60 nm, step 1. The ITU-T G.694.2 value is 20 nm. A value more than 1 nm away from 20 gives a warning.	20 nm
Custom wavelengths	Use the wavelengths typed in the table instead of the regular grid. Typing a value in the Custom wavelength column turns this on automatically.	Off; On.	Off
Custom wavelength per channel (nm)	Wavelength of each channel when custom wavelengths are on. Channels are sorted from the shortest wavelength.	400 to 1700 nm, step 0.01, one value per channel. Duplicate values are rejected.	1546.92 + 0.8 x (channel number - 1) nm

Parameter	What it controls	All possible values (dashboard limits)	Default
Bit rate per channel (Gb/s)	Data rate of each channel. The total nominal capacity is the sum of all channel rates.	0.1 to 100 Gb/s, step 0.5, set separately for each channel.	10 Gb/s
Launch power per channel (dBm)	Average optical power sent into the multiplexer by each laser.	-20 to 10 dBm, step 0.5, set separately for each channel.	0 dBm
MUX insertion loss (dB)	Loss of the multiplexer.	0 to 15 dB, step 0.5.	3 dB
DEMUX insertion loss (dB)	Loss of the demultiplexer.	0 to 15 dB, step 0.5.	3 dB
DEMUX 3 dB bandwidth / spacing	Pass-band width of each demultiplexer channel as a fraction of the distance to the nearest neighbour channel. A wider filter passes more neighbour light (crosstalk).	0.1 to 1.0, step 0.05.	0.5
DEMUX filter order	Steepness of the demultiplexer filter edges. Order 1 is Gaussian; a higher order has steeper edges.	1; 2; 3.	1
Isolation floor (dB)	Rejection of channels far from the pass-band. Sets the crosstalk floor.	10 to 70 dB, step 1.	40 dB

2.1 Grid checks made by the simulator

Table 5. WDM validation rules

Level	Condition
Error (simulation does not run)	Number of channels below 1 or above 16. Fewer custom wavelengths than channels, or a wavelength of zero or below. Two channels with the same wavelength (duplicates). A channel bit rate of zero or below. Minimum channel spacing smaller than the bit rate in Gb/s (for example 12.5 GHz at 25 Gb/s): the channel spectra overlap. CWDM with more than 18 channels.
Warning (simulation runs)	Minimum spacing below about twice the bit rate: strong crosstalk expected. CWDM spacing more than 1 nm away from 20 nm, or CWDM channels outside 1260 to 1625 nm. DWDM channels outside 1530 to 1625 nm. Amplifier on while the channels leave the 1528 to 1565 nm gain window. Channels outside the wavelength window of the selected fibre. DEMUX bandwidth narrower than about 1.2 times the bit rate (filter-induced inter-symbol interference is not modelled).
Information	CWDM start wavelength not on the ITU grid. DWDM spacing that is not 12.5, 25, 50, 100 or 200 GHz.

3. Transmitter and receiver (TxRxConfig)

Table 6. Transmitter and receiver parameters

Parameter	What it controls	All possible values (dashboard limits)	Default
Modulation format	Intensity modulation of the laser.	NRZ-OOK (non-return-to-zero on-off keying); RZ-OOK (return-to-zero, pulse in the first half of the bit).	NRZ-OOK
Extinction ratio (dB)	Ratio of the power of a 1 to the power of a 0.	3 to 25 dB, step 0.5.	10 dB
Tx rise time (fraction of bit)	10 to 90 percent rise time of the transmitter as a fraction of the bit period.	0.05 to 0.8, step 0.05.	0.25
Samples per bit	Time resolution of the waveforms.	8 to 64, step 8.	16
Bits displayed in waveforms	Length of the bit sequence used for the level and ISI statistics of the fast (analytic) evaluation of every channel. The waveform panels show the first 32 bits of the Monte-Carlo run.	128 to 1024, step 64.	256
Monte-Carlo BER bits	Length of the noisy time-domain run used for the simulated BER. A run with zero errors gives only an upper bound.	1000 to 200000, step 1000.	20000
Reproducible random seed (side bar)	Use a fixed seed so that the same settings always give the same bit pattern and noise.	Off; On.	On
Seed (side bar)	Value of the random seed.	0 to 1000000, step 1.	42
Decision threshold	How the receiver decides between 0 and 1.	Optimum (depends on the noise at both levels); Midpoint (halfway between the levels).	Optimum
Laser RIN (dB/Hz)	Relative intensity noise of the laser.	-170 to -110 dB/Hz, step 1.	-150 dB/Hz

Parameter	What it controls	All possible values (dashboard limits)	Default
Laser linewidth (MHz)	Optical linewidth. In intensity-modulated direct detection it affects the result only through spectral broadening and dispersion.	0.01 to 1000 MHz, step 1.	10 MHz
Source RMS spectral width (nm)	Spectral width of the source (for example from chirp). Together with the linewidth and the modulation it sets the chromatic-dispersion broadening.	0 to 5 nm, step 0.01.	0.03 nm
Responsivity (A/W)	Photodiode current per watt of light.	0.1 to 1.2 A/W, step 0.05. Typical values are about 0.9 for InGaAs at 1550 nm and 0.4 to 0.5 for silicon at 650 to 850 nm.	0.9 A/W
Receiver BW / bit rate	Receiver electrical 3 dB bandwidth as a fraction of the bit rate.	0.3 to 1.2, step 0.05.	0.7
Receiver filter	Shape of the receiver electrical filter.	Bessel (Gaussian, low ringing); Butterworth (maximally flat, more ringing).	Bessel
Butterworth order	Order of the Butterworth filter. Shown only when Butterworth is selected.	2; 3; 4; 5; 6.	4
Sensitivity (dBm)	Received power below which a channel is marked Degraded.	-50 to 0 dBm, step 1.	-24 dBm
Temperature (K)	Receiver temperature for thermal noise.	200 to 400 K, step 5.	300 K
Load resistance (ohm)	Equivalent load resistance for thermal noise.	10 to 10000 ohm, step 10.	500 ohm
Receiver noise figure (dB)	Extra noise of the receiver electronics.	0 to 15 dB, step 0.5.	3 dB
Dark current (nA)	Photodiode current in the dark.	0 to 1000 nA, step 1.	1 nA

Parameter	What it controls	All possible values (dashboard limits)	Default
RMS timing jitter (ps)	Random timing error of the sampling instant. Entered in the Noise tab, shown and used only when the Timing jitter switch is on.	0 to 100 ps, step 0.5.	2 ps

4. Noise configuration (NoiseConfig)

Every noise source has its own switch. A source that is switched off contributes exactly zero to the variance and its waveform panel shows that it is disabled.

Table 7. Noise switches

Switch	What it adds when on	All possible values	Default
Thermal noise	Random current of the receiver resistance and electronics.	On; Off.	On
Shot noise	Random current from the discrete arrival of photons and electrons. Grows with the received power.	On; Off.	On
Laser RIN noise	Intensity noise of the laser. Grows with the square of the received power.	On; Off.	On
Amplifier ASE noise	Amplified spontaneous emission of the optical amplifier, including signal-ASE and ASE-ASE beat noise. Has an effect only while the amplifier is on.	On; Off.	On (no effect while the amplifier is off)
Inter-channel crosstalk	Light of the other channels that leaks through the demultiplexer.	On; Off.	On

Switch	What it adds when on	All possible values	Default
Background optical noise	Extra broadband optical noise added at the receiver input. When on, its level can be set: -80 to -10 dBm per nm per polarisation, step 1.	On; Off.	Off. Level -45 dBm/nm (assumed).
Timing jitter	Random timing error of the sampling instant. When on, its size can be set in the Noise tab (0 to 100 ps, step 0.5).	On; Off.	Off. Size 2 ps.

5. Optical amplifier (AmpConfig)

Table 8. Amplifier parameters

Parameter	What it controls	All possible values (dashboard limits)	Default
Amplifier ON	Switch the optical amplifier (EDFA) on. When off, its gain and ASE noise are bypassed.	On; Off.	Off
Placement	Where the amplifier sits in the link.	Booster (right after the multiplexer); In-line (several amplifiers equally spaced along the fibre); Pre-amplifier (just before the demultiplexer, after any compensator).	Pre-amplifier
Gain (dB)	Requested gain of each amplifier.	0 to 40 dB, step 0.5.	15 dB
Noise figure (dB)	Noise figure of each amplifier. A value below the physical (quantum) limit for the chosen gain gives a warning.	3 to 12 dB, step 0.5.	5 dB

Parameter	What it controls	All possible values (dashboard limits)	Default
In-line amplifiers (count)	Number of in-line amplifiers. The fibre is divided into one more equal spans. Used only for the In-line placement.	1 to 8, step 1.	1
Output power limit (saturation)	Limit the total output power of an amplifier. The gain is reduced so that the output does not exceed the limit, but never below 0 dB.	On; Off.	On
Total output power limit (dBm)	The limit for the total power of all channels.	0 to 30 dBm, step 0.5.	17 dBm

6. Dispersion and compensation (DispConfig)

Table 9. Dispersion parameters

Parameter	What it controls	All possible values (dashboard limits)	Default
Fibre dispersion ON	Include chromatic, modal and polarisation-mode dispersion. When off, there is no pulse broadening.	On; Off.	On
Dispersion compensation ON	Reduce the chromatic part of the broadening. Modal dispersion and PMD are not compensated.	On; Off.	Off
Method	Type of compensator.	DSP (digital signal processing, no extra optical loss); DCF (dispersion-compensating fibre module, adds insertion loss).	DSP
Compensated fraction of chromatic broadening	Share of the chromatic broadening that is removed.	0 to 1, step 0.05 (0 = nothing, 1 = all).	0.9

Parameter	What it controls	All possible values (dashboard limits)	Default
DCF insertion loss (dB)	Loss of the DCF module. Applies only when the method is DCF and compensation is on.	0 to 15 dB, step 0.5.	5 dB (assumed)

WDM itself does not remove dispersion: each wavelength is dispersed separately.

7. Network traffic (TrafficConfig)

Traffic balancing is a network-level simulation. There is no physical optical switching fabric, so traffic can move only between channels of the same compatibility group.

Table 10. Traffic parameters

Parameter	What it controls	All possible values (dashboard limits)	Default
Demand per channel (Gb/s)	Traffic offered to each channel.	0 to 200 Gb/s, step 0.5, set separately for each channel.	9.5; 6; 3; 8.8; 2; 5; 4; 7 Gb/s (channels 1 to 8; the list repeats for channels 9 to 16)
Priority (1 = high)	Importance of the traffic of each channel. Used when traffic is moved or dropped.	1; 2; 3; 4; 5 (1 is the highest priority).	1; 2; 3; 1; 3; 2; 3; 2 for channels 1 to 8
Group	Compatibility group. Traffic can be moved only between channels with the same group number.	1 to 8.	1 for every channel
Congestion threshold	Fraction of the capacity above which a channel is congested and traffic is moved away.	0.3 to 1.0, step 0.05.	0.8
Adaptive load balancing	Move traffic above the threshold to channels of the same group that still have room.	On; Off. When off, the original allocation is kept.	On
Priority scheduling	When a channel is overloaded, drop the lowest-priority traffic first. When off, the loss is shared in proportion.	On; Off.	On
Re-route latency (us)	Extra delay added to traffic that was moved to another channel (assumed).	0 to 1000 us, step 10.	50 us

Parameter	What it controls	All possible values (dashboard limits)	Default
Packet size (bytes)	Packet length used for the packet error rate and the latency. Not shown in the dashboard; set it in a JSON configuration.	Any positive whole number of bytes.	1500 bytes

8. Channel failure and recovery (FaultConfig)

All times in this section are simulated values calculated from the settings. They are not measurements of real switching hardware.

Table 11. Fault parameters

Parameter	What it controls	All possible values (dashboard limits)	Default
Failed channels	Channels whose laser is lost. Set with the Trigger failure and Restore failed channels buttons.	Any subset of channels 1 to N (N = number of channels), including all of them.	None
Channel to fail	Channel that the Trigger failure button disables.	1 to N.	1
Protection switching	Re-route the traffic of a failed channel automatically.	On; Off.	On
Backup route	Where the traffic of a failed channel goes.	Reroute to remaining channels (same group, up to the congestion threshold); Dedicated backup channel (one chosen channel, up to its full capacity, then the remaining channels if needed).	Reroute to remaining channels
Backup channel	The channel used when the dedicated backup route is chosen. It cannot be a failed channel.	1 to N.	Channel 7 in the dashboard (channel 8 if the stored configuration value 7 is used, because it counts from 0).
Failure time (ms)	Moment on the timeline at which the failure happens.	0 to 10000 ms, step 10.	100 ms
Monitor poll period (ms)	How often the power monitor checks each channel.	1 to 1000 ms, step 1.	10 ms
Debounce (polls)	Number of consecutive bad readings needed before a failure is declared.	1 to 20, step 1.	3

Parameter	What it controls	All possible values (dashboard limits)	Default
Switch time (ms)	Time of the protection switch after detection (assumed value).	0 to 5000 ms, step 5.	50 ms
Timeline end (ms)	Length of the plotted timeline. It is extended automatically if recovery finishes later.	100 to 20000 ms, step 50.	400 ms

Detection time = (whole number of (failure time / poll period) + debounce) x poll period. Recovery time = detection time + switch time, measured from the failure time. With the defaults, detection happens 30 ms after the failure and recovery after 80 ms.

9. Other dashboard controls

Table 12. Controls in the side bar

Parameter	What it controls	All possible values (dashboard limits)	Default
Run Simulation	Apply the current settings and recalculate everything.	Button.	Not applicable
Auto-update when settings change	When on, the results update after every change. When off, they change only after Run Simulation.	On; Off.	On
Reset to Defaults	Restore every setting of every section to its default value.	Button.	Not applicable
Channel inspected in detail	Channel shown in the waveform, noise, eye and dispersion views.	1 to N. If that channel has failed, the first healthy channel is shown and a message explains it.	1

10. All selectable choices at a glance

Table 13. Every drop-down, switch and option list

Setting	Choices	Number of choices
Fibre material	Silica; POF; PCS	3
Fibre structure	Single-mode; Multimode	2
Refractive-index profile	Step-index; Graded-index	2
Complete fibre combinations	3 x 2 x 2	12
WDM scheme	DWDM; CWDM	2
DWDM frequency spacing	12.5; 25; 50; 100; 200 GHz	5
DEMUX filter order	1; 2; 3	3
Modulation format	NRZ-OOK; RZ-OOK	2
Decision threshold	Optimum; Midpoint	2
Receiver filter	Bessel; Butterworth	2
Butterworth order	2; 3; 4; 5; 6	5
Noise switches (each on or off)	Thermal; Shot; Laser RIN; Amplifier ASE; Crosstalk; Background; Timing jitter	7 switches
Amplifier placement	Booster; In-line; Pre-amplifier	3
Dispersion compensation method	DSP; DCF	2
Traffic switches	Adaptive load balancing; Priority scheduling (each on or off)	2 switches
Traffic priority	1; 2; 3; 4; 5	5
Traffic group	1 to 8	8
Backup route	Reroute to remaining channels; Dedicated backup channel	2

Setting	Choices	Number of choices
Other switches	Protection switching; Amplifier ON; Fibre dispersion ON; Dispersion compensation ON; Custom wavelengths; Manual attenuation; Manual dispersion; Operating wavelength follows channels; Reproducible seed; Auto-update	10 switches